

Smart Radon Fan - Industry Alignment and Impact

Executive Summary

Modern radon mitigation systems focus primarily on confirming that a fan is operating, not that indoor radon levels remain safe over time. Smart Radon Fan introduces continuous performance monitoring and optimization while preserving compliance with existing mitigation standards. Rather than replacing current practices, it strengthens their core objective: Long-term protection of occupants from radon exposure.

Current Industry Practice

Typical mitigation systems rely on Active Soil Depressurization (ASD) combined with a manometer indicating suction. While effective at installation, these systems provide limited ongoing assurance that radon levels remain reduced. Homeowners rarely perform regular follow-up testing, leaving a gap between mechanical operation and verified safety.

The Industry Gap

ANSI/AARST standards require operational monitoring and failure notification but do not require continuous radon verification. Existing systems confirm airflow rather than health outcome, meaning elevated radon levels may go unnoticed for extended periods.

Role of Smart Radon Fan

Smart Radon Fan functions as a performance monitoring and optimization system for an already compliant mitigation installation. It maintains traditional mitigation hardware while adding continuous radon monitoring, automated fan optimization, and proactive notifications when unhealthy conditions persist.

Traditional Mitigation vs Smart Radon Fan

Traditional Mitigation	Smart Radon Fan
Fan operation assumed effective	Radon reduction continuously verified
Passive visual monitor	Active automated notifications
Periodic homeowner testing	Continuous monitoring and alerts
Mechanical assurance	Health-outcome assurance

Key Innovation: Outcome-Based Monitoring

Traditional mitigation answers the question: Is the fan running? Smart Radon Fan answers the question: Is the home actually protected?

Using continuous data monitoring and fan control, and including radon level notifications, the system controls the radon fan and alerts homeowners when long-term radon levels remain elevated, transforming mitigation into an actively supervised environmental safety system under automated control.

Alignment with Industry Standards

Smart Radon Fan does not replace existing mitigation requirements. Instead, it enhances them by supporting ongoing monitoring expectations, enabling remote notifications, and providing long-term operational documentation aligned with OM&M practices.

Conclusion

Smart Radon Fan represents a natural evolution of radon mitigation from install-and-forget infrastructure to continuously verified occupant protection. By augmenting compliant systems rather than disrupting them, it advances industry practice while remaining fully compatible with established standards.